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UGC AUTONOMOUS INSTITUTION
Maisammaguda (V), Kompally - 500100, Secunderabad, Telangana State, India
UGC - Autonomous Institute
Accredited by NBA & NAAC with 'A' Grade
Approved by AICTE
Permanently affiliated to JNTUH

Project Based Learning

On

COMBINED FOOTING DESIGN USING AUTOCAD

By

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ABSTRACT

Combined footing is a reinforced cement concrete foundation provided to support two or more columns when individual isolated footings cannot be provided conveniently due to space limitations or the close spacing of columns. Proper design and detailing of a combined footing ensure safe transfer of structural loads from the columns to the supporting soil.

This project presents the design and drafting of a rectangular RCC combined footing using AutoCAD. The drawing includes the footing plan, sections, isometric view, column and pedestal details, dimensions, and schedule of structural elements.

The project focuses on preparing accurate 2D engineering drawings with proper dimensions and clear structural representation. The combined footing considered in the drawing has a footing size of 3500 × 2500 mm, with a 600 mm footing thickness, 600 × 600 mm RCC pedestals, and 450 × 450 mm RCC columns.

AutoCAD is used to create precise plans, sections, dimensions, annotations, and structural details.

INTRODUCTION

A combined footing is a type of foundation used to support two or more columns on a common footing slab. It is generally adopted when columns are closely spaced or when the location of a column is near a property boundary, making a separate isolated footing unsuitable.

The main purpose of a combined footing is to safely transfer the loads from the supported columns to the soil while maintaining adequate stability and minimizing excessive settlement. Proper proportioning and detailing of the footing are therefore important for the safety and durability of the structure.

This project deals with the design and drafting of a rectangular combined footing using AutoCAD. The drawing represents two RCC columns supported on a common RCC footing through individual pedestals.

The drawing includes the plan, Section X-X', Section Y-Y', and isometric view of the combined footing. It also provides dimensions and a schedule of structural elements for the columns, pedestals, footing, and PCC.

The AutoCAD drawing provides an accurate representation of the foundation arrangement and helps engineers and students understand the relationship between the footing, pedestal, column, and PCC layers.

This project focuses on the design and drafting of a rectangular RCC combined footing using AutoCAD. The drawing represents two RCC columns supported by individual pedestals over a common footing, with a PCC layer provided below the footing.

DRAWING

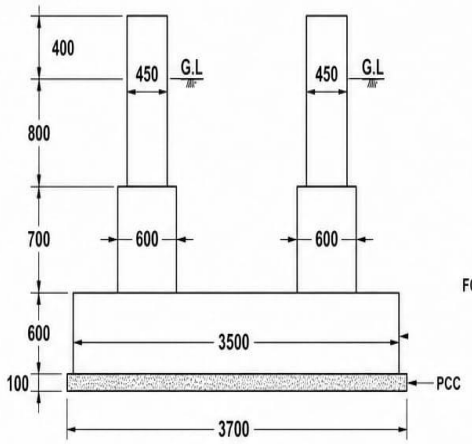
COMBINED FOOTING DSIGN

COMBINED FOOTING DESIGN – PLAN DETAILS

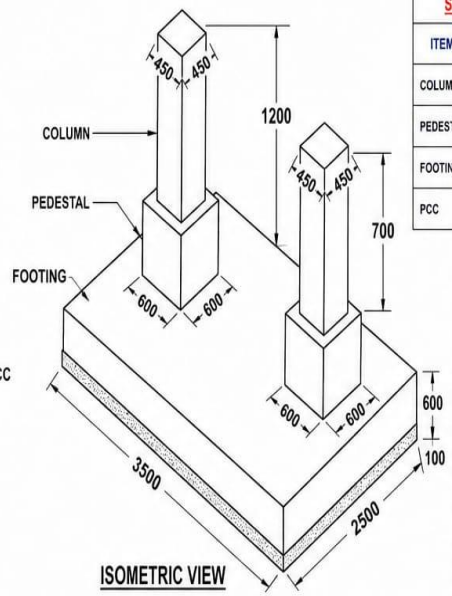
Description	**Size / Details**
Footing Type	RCC Combined Footing
Footing Length	3500 mm
Footing Width	2500 mm
Footing Thickness	600 mm
PCC Length	3700 mm
PCC Width	2700 mm
PCC Thickness	100 mm
Pedestal Size	600 × 600 mm
Column Size	450 × 450 mm

Plan Includes

- RCC combined footing
- Two RCC columns
- Two RCC pedestals
- PCC layer
- Column identification as C1 and C2
- Footing dimensions
- PCC dimensions
- Column and pedestal dimensions
- Centre/reference lines
- Structural element locations



SECTION X-X'



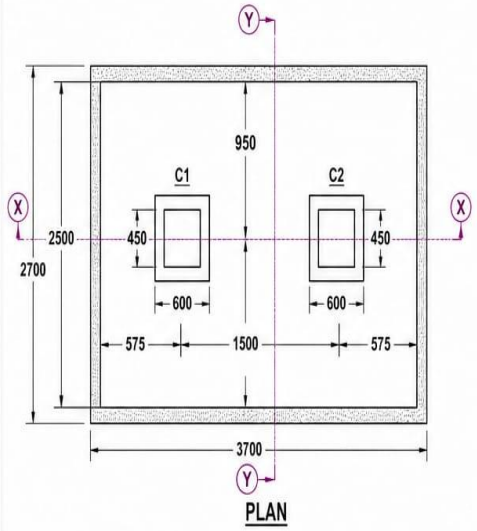
ISOMETRIC VIEW

SCHEDULE OF STRUCTURAL ELEMENTS			
ITEM	DESCRIPTION	SIZE	REMARKS
COLUMNS	RCC COLUMN	450 X 450	HEIGHT = 1200
PEDESTAL	RCC PEDESTAL	600 X 600	HEIGHT = 700
FOOTING	RCC FOOTING	3500 X 2500	THICKNESS = 600
PCC	PCC (1:4:8)	3700 X 2700	THICKNESS = 100

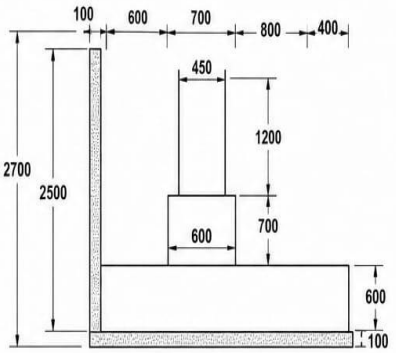
SCHEDULE OF DIMENSIONS (mm)	
DESCRIPTION	SIZE (mm)
FOOTING (L X B)	3500 X 2500
FOOTING THICKNESS	600
PCC (L X B)	3700 X 2700
PCC THICKNESS	100
PEDESTAL (L X B)	600 X 600
PEDESTAL HEIGHT	700
COLUMN (L X B)	450 X 450
COLUMN HEIGHT (ABOVE PEDESTAL)	1200
TOTAL HEIGHT (FROM TOP OF PCC)	2400

GENERAL NOTES :

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DO NOT SCALE THE DRAWING.



PLAN



SECTION Y-Y'



DIMENSION CALCULATIONS

1. PCC Length

Footing length = 3500 mm

PCC projection = 100 mm on each side

Therefore,

PCC Length = 3500 + 100 + 100

PCC Length = 3700 mm

2. PCC Width

Footing width = 2500 mm

PCC projection = 100 mm on each side

Therefore,

PCC Width = 2500 + 100 + 100

PCC Width = 2700 mm

3. Footing Thickness

Footing Thickness = 600 mm

4. Pedestal Size

Pedestal = 600 × 600 mm

5. Column Size

Column = 450 × 450 mm

6. Vertical Structural Dimensions

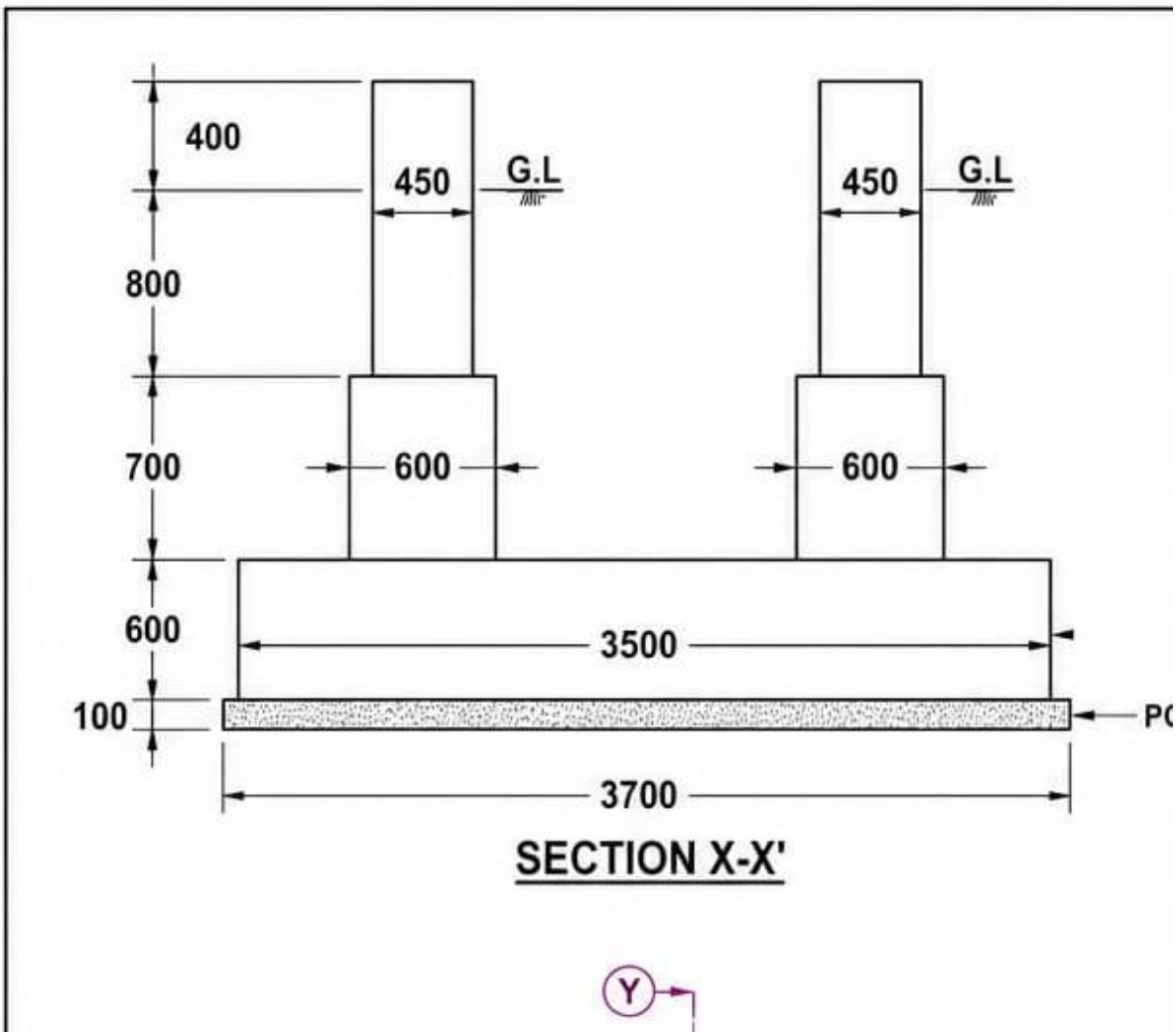
From the drawing:

Footing thickness = 600 mm

Pedestal height = 700 mm

Column height above pedestal = 1200 mm

DRAWING 2



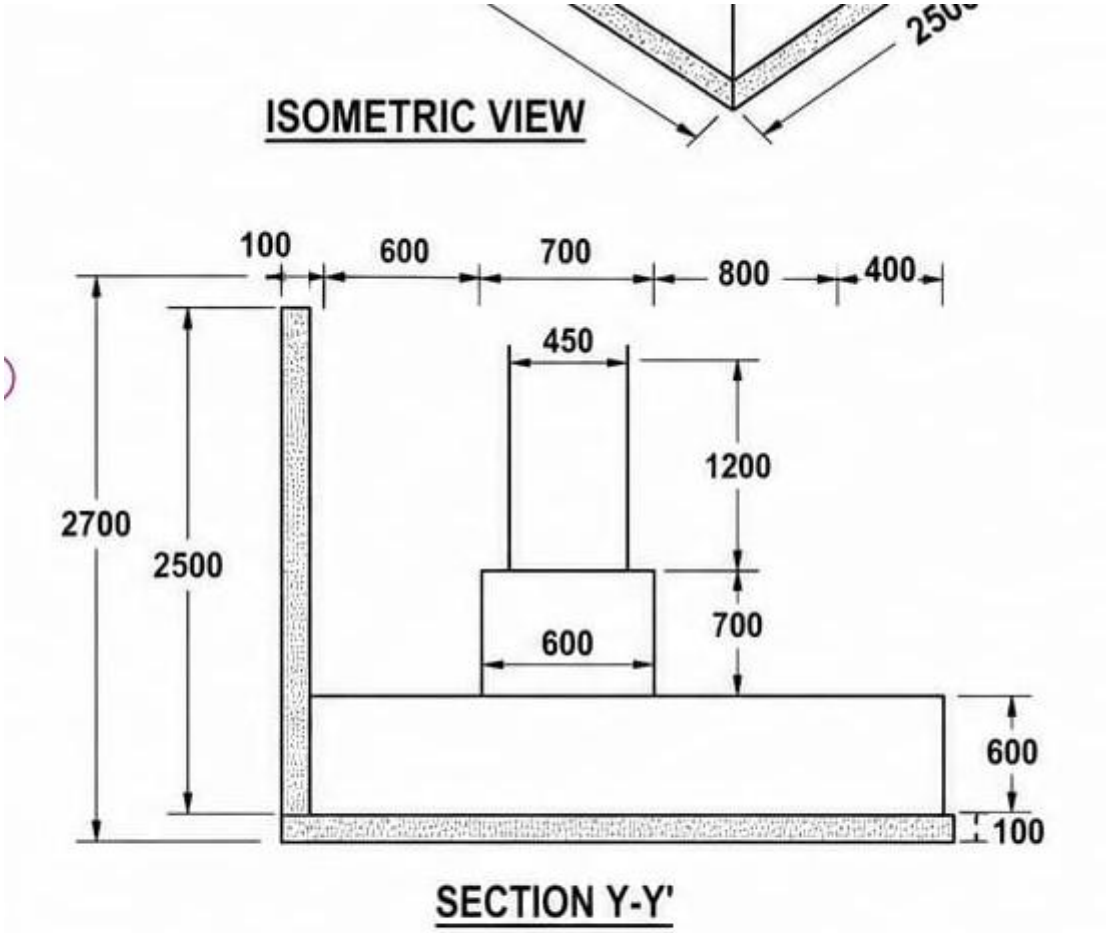
Section Details

Section X-X' represents the vertical cross-section of the combined footing along the column arrangement.

Components Shown

- RCC combined footing
- RCC pedestals
- RCC columns C1 and C2
- PCC layer
- Ground level
- Footing thickness
- Pedestal height
- Column height
- Overall foundation dimensions

DRAWING 3


ISOMETRIC VIEW
100
600
700
800
400
450
1200
2700
2500
700
600
600
100
SECTION Y-Y'

SECTION Y-Y'

Section Y-Y' shows the transverse section of the combined footing and illustrates the relationship between the PCC, footing, pedestal, and column.

Section Includes

- RCC combined footing
- PCC layer
- RCC pedestal
- RCC column
- Footing thickness
- Pedestal dimensions
- Column dimensions
- Ground/reference level
- Vertical dimensions

The section provides a clear understanding of the depth and vertical arrangement of the foundation components.

ISOMETRIC VIEW

The isometric view provides a three-dimensional representation of the combined footing.

It shows:

- RCC combined footing
- Two RCC pedestals

- Two RCC columns
- PCC layer
- Footing dimensions
- Pedestal dimensions
- Column dimensions

The isometric view helps in understanding the overall shape and arrangement of the foundation before construction.

DRAWING DETAILING WITH MEASUREMENTS

Foundation

- Footing Length – 3500 mm
- Footing Width – 2500 mm
- Footing Thickness – 600 mm
- PCC Length – 3700 mm
- PCC Width – 2700 mm
- PCC Thickness – 100 mm

Pedestal

- Pedestal Size – 600 × 600 mm
- Pedestal Height – 700 mm

Columns

- Column Size – 450 × 450 mm
- Column Height Above Pedestal – 1200 mm
- Number of Columns – 2

AUTOCAD DRAWING TOOLS USED

Drawing Tools

- Line
- Polyline
- Rectangle
- Circle
- Arc
- Hatch
- Text
- MText
- Dimension
- Leader

Modify Tools

- Move
- Copy
- Offset
- Mirror
- Trim
- Extend
- Rotate
- Scale
- Stretch
- Array
- Fillet
- Chamfer
- Explode

TECHNOLOGY USED

Software

AutoCAD 2024

Operating System

Windows 10 / Windows 11

Drafting Method

2D Computer-Aided Design (CAD)

AutoCAD is used for preparing the combined footing plan, sections, isometric representation, dimensions, annotations, and structural schedules.

DESIGN STANDARDS

The combined footing drawing should be prepared with reference to applicable Indian Standards and standard reinforced-concrete design practices.

- **IS 456:2000** – Plain and Reinforced Concrete
- **IS 1904** – Design and Construction of Foundations in Soils
- **IS 2502** – Code of Practice for Bending and Fixing of Bars for Concrete Reinforcement
- **SP 34** – Handbook on Concrete Reinforcement and Detailing

Note: The supplied drawing itself does not specify reinforcement bar sizes or spacing, so reinforcement values should not be added unless they are provided separately or calculated as part of the structural design.

APPLICATIONS

Combined footings are commonly used in:

- Residential buildings
- Commercial buildings
- Apartment buildings
- Industrial structures
- Institutional buildings
- Office buildings
- Boundary-column foundations
- Closely spaced column foundations
- RCC framed structures

ADVANTAGES

- Supports multiple columns using a common foundation
- Suitable for closely spaced columns
- Useful where isolated footing dimensions are restricted
- Provides efficient load transfer to the soil
- Helps maintain proper structural stability
- Reduces interference between adjacent footings
- Provides accurate structural representation using AutoCAD
- Easy to modify and revise during the design stage
- Produces professional engineering drawings
- Improves construction communication

CONCLUSION

The project demonstrates the design and drafting of a rectangular RCC combined footing using AutoCAD. The drawing includes the combined footing plan, Sections X-X' and Y-Y', isometric view, column and pedestal details, PCC arrangement, dimensions, and structural element schedule.
The combined footing considered in the project consists of a

3500 × 2500 mm RCC footing with 600 mm thickness, supported by 600 × 600 mm pedestals and 450 × 450 mm RCC columns. A 3700 × 2700 mm PCC layer with 100 mm thickness is provided below the footing as shown in the drawing.
The use of AutoCAD enables accurate preparation of plans, sections, dimensions, annotations, and structural details. The completed drawing provides a clear engineering reference for understanding and presenting the combined footing arrangement and can be useful for academic learning and civil engineering drafting applications.